DIGITAL CLOCK

A PROJECT REPORT

Submitted by

(YOGESWARAN M) (8115U24ME062)

in partial fulfillment of requirements for the award of the course

MGB1201 - PYTHON PROGRAMMING

in

DEPARTMENT OF

MECHANICAL ENGINEERING

Under the Guidance of

S.RAJESWARI

Department of Computer Science and Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY– 621112.

ANNA UNIVERSITY

CHENNAI 600025.

DECEMBER - 2025

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Under

ANNA UNIVERSITY, CHENNAI

K. RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)

SAMAYAPURAM – 621 112

BONAFIDE CERTIFICATE

Certified that this project report on **"Digital Clock"** is the bonafide work of **(YOGESWARAN M)** who carried out the project work during the academic year 2025 - 2026 under my supervision.

Dr. R. SASIKUMAR,M.E., Ph.D.
HEAD OF THE DEPARTMENT
ASSISTANT PROFESSOR
Department of Computer Science and Engineering,
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy

Mrs. RAJESHWARI.,M.E
SUPERVISOR
ASSISTANT PROFESSOR
Department of Computer Science and Engineering,
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy

Submitted for the viva-voce examination held on……

SIGNATURE OF INTERNAL EXAMINER

SIGNATURE OF EXTERNAL EXAMINER

DECLARATION

I declare that the project report on **"(Digital Clock)"** is the result of original work done by me and best of my knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of **BACHELOR OF ENGINEERING**. This project report is submitted on the partial fulfilment of the requirement of the completion of the course **MGB1201 – PYTHON PROGRAMMING.**

Signature

_____

(YOGESWARAN M)

Place: Samayapuram

Date:

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I wish to express my special thanks to the officials and Lab Technicians of my department who rendered their help during the period of the work progress.

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MISSION OF THE INSTITUTION

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ABSTRACT

A digital clock is a type of clock that displays the time in numeric format using digits or other symbols. In this project, we will create a digital clock using Python, a popular programming language known for its simplicity and versatility. The clock will display the current time in hours, minutes, and seconds, and will be updated in real-time. We will use Python's Tkinter library to create a graphical user interface (GUI) for the clock, and the time module to get the current time. The clock will also have features such as alarm and stopwatch, making it a useful tool for everyday life. By creating this digital clock, we will demonstrate the power and flexibility of Python programming, and show how it can be used to create practical and user-friendly applications. The digital clock will be a great example of how Python can be used to create real-world applications that are both functional and visually appealing. With its simple and intuitive interface, the digital clock will be easy to use and understand, making it a great project for beginners and experienced programmers alike. Overall, this project will showcase the capabilities of Python and provide a useful tool for anyone looking to create a digital clock.

ABSTRACT WITH POs AND PSOs MAPPING

CO 5 : BUILD PYTHON APPLICATIONS FOR SOLVING REAL-TIME PROBLEMS

ABSTRACT	POs MAPPED	PSOs MAPPED
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Note: 1- Low, 2-Medium, 3- High

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CHAPTER 1

INTRODUCTION

A digital clock is a type of clock that displays the time in numeric format using digits or other symbols. In this project, we will create a digital clock using Python, a popular programming language known for its simplicity and versatility. The clock will display the current time in hours, minutes, and seconds, and will be updated in real-time. We will use Python's Tkinter library to create a graphical user interface (GUI) for the clock, and the time module to get the current time. The clock will also have features such as alarm and stopwatch, making it a useful tool for everyday life. By creating this digital clock, we will demonstrate the power and flexibility of Python programming, and show how it can be used to create practical and user-friendly applications. The digital clock will be a great example of how Python can be used to create real-world applications that are both functional and visually appealing. With its simple and intuitive interface, the digital clock will be easy to use and understand, making it a great project for beginners and experienced programmers alike. Overall, this project will showcase the capabilities of Python and provide a useful tool for anyone looking to create a digital clock.

1.1 Objective

The primary objective of the Python-based Titli Digital Clock is to design and implement a responsive, visually appealing, and accurate desktop clock application using only basic Python modules, serving as a robust educational exercise in GUI and time management. This objective emphasizes simplicity, utilizing standard library features to avoid external dependencies.

1.1.1 Core Timekeeping Objective

The fundamental goal is to continuously retrieve the system's current time and display it with precision. This is achieved through the time module's strftime() function, which converts the raw system timestamp into a human-readable string. The core objective is represented by the continuous loop that ensures the displayed time never lags behind.

```
import time

def get_formatted_time():

    The strftime function formats the current system time

    %H: 24-hour clock, %M: minute, %S: second, %p: AM/PM (if needed)

    current_time_str = time.strftime('%H:%M:%S')

    return current_time_str
```

1.1.2. GUI and User Experience Objective

A key objective is to create a clean, responsive Graphical User Interface (GUI). We employ the tkinter module, which is standard with most Python distributions, to create the main window and display the time using a Label widget. The display must be clear, using a large font and high-contrast colors to enhance readability. This involves setting the widget properties such as font, background, and foreground. The goal is for the application window to act as a stand-alone digital clock that is highly visible on the desktop.

1.1.3. Real-Time Update Objective

The most critical objective in a clock application is to ensure the display updates every second without freezing the entire program. This is handled using tkinter's built-in after() method, which schedules a function to run after a specified delay (typically 1000 milliseconds for a one-second update). This method leverages Python's event loop to manage the timing, making the application responsive and efficient.

Time Update Interval=1000 milliseconds

The project thus serves to cement understanding of module imports, function definition, string formatting, and event-driven GUI handling—all foundational pillars of Python programming.

1.2 Overview

The Overview of the Titli Digital Clock project details the three major components and the sequential logic that drives the application, all built upon basic Python constructs.

1.2.1. The Graphical Interface Component (Tkinter)

The clock's interface is initiated by creating the root window using root = tk.Tk(). This object forms the foundation of the GUI. A Label widget is then created within this root window to serve as the sole display area for the time string. The label is styled for visual appeal:

Python

```
# Initialization of the main GUI components
root = tk.Tk()
root.title('Titli Digital Clock')
# The Label widget will hold the time string
clock_label = tk.Label(root, font=('Arial', 60, 'bold'), bg='black', fg='cyan')
clock_label.pack(anchor='center') # Place the label in the center
```

This structure is an excellent example of widget hierarchy in GUI programming. The pack() method is used to manage the layout, ensuring the label remains centered within the window.

1.2.2. The Time Logic Component (time module)

The time module provides the critical function for accessing the host system's time. The key is the time.strftime(format_string) method. This method takes format codes (like %H, %M, %S) and returns a formatted string. This abstraction allows the Python application to rely on the operating system for accurate timekeeping, bypassing complex hardware clock management.

```
from time import strftime
```

```
def update_time():
    Retrieve current time from system and format it
    current_time = strftime('%I:%M:%S %p')
    Update the Label text with the new time string
    clock_label.config(text=current_time)
    Recursively schedule the function to run again
    clock_label.after(1000, update_time)
```

1.2.3. The Event Loop and Real-Time Update Mechanism

The application's real-time capability is achieved by combining the update_time() function and the after() method. The after(1000, update_time) line is the core mechanism; it tells the tkinter main loop to call the update_time function again after a 1000-millisecond (1-second) delay. The entire application is kept alive by the root.mainloop() call, which initiates the event loop, continuously monitoring for events (like the after timer expiring) and keeping the window open. This pattern demonstrates recursion and event scheduling—critical concepts for any real-time application.

1.3 Technology used

The Python-based Titli Digital Clock exclusively relies on core Python technologies that are built into the standard distribution, highlighting the platform's ability to handle complex tasks with minimal external dependencies.

1.3.1. Programming Language: Python 3

The entire project is written in Python 3, chosen for its clear syntax and robust standard library. Python simplifies the logic flow, making the code highly readable and easy to maintain. Key language features used include:

Function Definitions (def): For modularity, the core logic for fetching and updating the time is encapsulated in a dedicated function, update_time().

String Formatting: Essential for display, the time.strftime() method uses format codes (e.g., %H, %M, %S) to structure the output string perfectly. This showcases Python's powerful string manipulation capabilities.

Variable Assignment: Variables like root (the main window) and clock_label (the display element) are instances of classes, demonstrating Object-Oriented Programming (OOP) fundamentals where objects hold specific properties and methods.

1.3.2. Core Module 1: tkinter (The GUI Toolkit)

The tkinter module is the sole graphical technology used. It is a wrapper around the Tcl/Tk toolkit and is standard on most systems.

Widget Implementation: The project uses the Tk() class for the main window and the Label class to display the time.

Method Usage:

config(): Used dynamically to update the text of the Label widget in real-time, demonstrating how to modify object properties during runtime.

after(delay, callback): The core scheduling mechanism. This function is fundamental to creating the refresh loop, scheduling the update_time function to execute again after the specified delay.

mainloop(): Initiates the event loop, which listens for the scheduled after event and keeps the application responsive.

1.3.3. Core Module 2: time (The Time Library)

The time module is the source of all time data.

strftime(): The most critical function, converting the system's raw time into a formatted string. For example, using the format string '\%I:\%M:\%S \%p' ensures the time is displayed in the standard 12-hour format with AM/PM designation.

CHAPTER 2

PROJECT METHODOLOGY

2.1 Problem Description

Problem Definition (Python Basics)

The Titli Digital Clock, implemented using Python's tkinter and time modules, is designed to overcome four primary functional and usability deficiencies present in basic, monolithic software implementations. These problems range from the non-trivial task of maintaining absolute temporal accuracy to ensuring the clock remains a responsive and feature-rich utility within a modern computing environment.

2.1. Time Accuracy

In a Python application, the problem of Time Accuracy shifts from hardware-based quartz crystal drift to software-based synchronization, network latency, and the inherent limits of the host operating system's clock. The primary problem is the lack of a native, built-in mechanism for Python to self-correct against a definitive global time standard, relying instead on the potentially drifting host clock. While Python's time.time() function returns the number of seconds since the Unix epoch (Jan 1, 1970), this value is drawn directly from the operating system's internal clock. If the host system's hardware clock is inaccurate (e.g., due to poor internal oscillator quality or infrequent OS-level synchronization), the Python clock will be inherently inaccurate, propagating the error.

The Non-Monotonic Problem: The standard time.time() clock is adjustable

(adjustable: True in time.get_clock_info('time')). This means the value can jump forward or backward due to OS-level time corrections (e.g., a time server update or a manual user change). This creates non-monotonicity, where time doesn't always move strictly forward, which can cause significant issues for logging, sequencing events, and accurate time calculation within the Titli Clock's logic.

The Python Solution Contrast: The project must explicitly utilize the time.monotonic() function (or the datetime module's robust capabilities) for calculating intervals (like setting an alarm delay). The time.monotonic() clock is designed to never go backward, solving the non-monotonic problem, even if the system clock itself is corrected. This is a critical distinction that must be programmed deliberately.

2.1.2. Low Readability

The issue of low readability can be simulated using Python string formatting and conditional logic to model how the clock's output changes based on "ambient conditions." The visual output of a digital clock is essentially a formatted string representation of time (e.g., "12:34:56"). We can use Python's print() function and various string methods (like str.format() or f-strings) to represent the display. To simulate low contrast in bright light, we might use variables to represent the display brightness and ambient light intensity. If the ratio of brightness to ambient light falls below a certain threshold (modeled by an if statement), we can use different character sets (e.g., printing all lowercase vs. all uppercase characters, or using special Unicode characters) to visually represent the time, illustrating the degradation in clarity. To address the problem of excessive brightness at night, we use another conditional check. If a variable representing the ambient light sensor is below a set nighttime threshold, the program could switch to a different output format, such as dimming the displayed numbers by printing spaces between digits or using a less intrusive colour code (if the terminal supports it).

Furthermore, basic readability checks can be implemented using a function that takes the time string and the current light level as arguments, returning a Boolean (True or False) indicating if the time is readable based on predetermined criteria. This exercise relies heavily on string manipulation, integer/float comparison, and function definition to abstract the hardware problem into a logical, solvable computational challenge.

2.1.3 Cannot Obey Modern Technology

The problem of the clock's isolation, or its inability to obey modern technology, can be abstractly modeled in Python by simulating command parsing and remote function calls. Without actual network connectivity, we can use dictionary data structures to represent the clock's internal state (time, alarm setting, time zone) and model incoming commands as string inputs. A remote control function, say set_alarm(time_str, is_on), can be defined to modify the clock state dictionary. The core challenge of connecting to modern technology is the ability to receive and interpret standardized instructions. This can be simulated using input() to receive a command string (e.g., "CMD:ALARM_SET|06:30|TRUE"). We then use string methods like str.split() to parse this command into its constituent parts (command name, time, status). Conditional statements (if/elif/else) are essential here to route the parsed command to the correct internal function (e.g., if command_name == "ALARM_SET": ...). This demonstrates the fundamental logic of an API or network listener: taking a flexible input, validating it, extracting necessary parameters, and executing a state change. Furthermore, a simple list can be used to store a queue of commands, illustrating how a connected device handles multiple instructions. This Python exercise primarily utilizes functions, string parsing, dictionaries for state management, and control flow to illustrate the core logical requirement for modern digital clock connectivity.

2.1.4 Less Additional Features

The lack of additional features can be solved by expanding the clock's internal data storage and output logic using Python's capabilities. A traditional clock only needs to store a few variables (hour, minute, alarm time). A feature-rich clock requires more complex data structures. For example, to provide a weather forecast feature, we could use a list of dictionaries to hold a multi-day forecast, where each dictionary contains keys like 'day', 'temp', and 'condition'. The clock's display logic would then use a for loop to iterate through this list and display each day's forecast sequentially. For enhanced alarm features, we can define a class called Alarm. This class could hold attributes (using variables) not just for the time, but for the desired wake-up music file (music_file), the volume level (volume), and a boolean flag for a 'smart wake' feature (smart_mode). An object instantiated from this class would represent a powerful, customized alarm. The core of this solution lies in defining and manipulating these complex data structures (lists, dictionaries, or custom classes) to hold and process information beyond simple timekeeping. The function responsible for displaying the time now has to check these multiple data structures and use string concatenation or f-strings to present a complex, aggregated display (e.g., Time + Temperature + Next Alarm). This demonstrates how Python can model the required data organization and logic flow for turning a basic time teller into an information hub.

2.2 Proposed Work – Digital Clock

The proposed work defines the development of the Titli Digital Clock as a robust, feature-rich desktop application built exclusively using Python's standard libraries, primarily focusing on tkinter for the GUI and time/datetime for time logic. This section outlines the functional features and the methodological approach to ensuring system integrity and temporal accuracy.

2.2.1 Accurate Time Keeping

The goal is to maintain and display time with system-level accuracy by correctly implementing Python's time functions within a non-blocking GUI loop.

Core Time Acquisition and Formatting

The proposed methodology centers on using the time.strftime() function. The work will detail how this function accesses the host operating system's (OS) highly accurate internal clock. The content will explain the Format Codes (e.g., %H, %M, %S, %p) and how they are used to render the time string. The system must update every second without freezing the GUI. This is achieved exclusively through the tkinter.after() method. The content will dedicate significant space to contrasting the correct, non-blocking use of tkinter.after(1000, update_function) with the incorrect, blocking use of time.sleep(1) inside a main loop, explaining why the latter locks the Python Global Interpreter Lock (GIL), preventing the GUI from updating.

2.2.2 GUI Framework

The graphical user interface (GUI) is an essential part of the digital clock. We will use the Tkinter library, a built-in Python library, to create the GUI. Tkinter provides a simple and easy-to-use interface for creating GUIs, making it ideal for beginners and experienced programmers alike. The GUI will consist of a window that displays the current time, with options to set alarms and customize the clock's appearance.

The GUI framework will include the following components:

1. A label to display the current time
2. Buttons to set alarms
3. A menu to customize the clock's appearance
4. A status bar to display messages

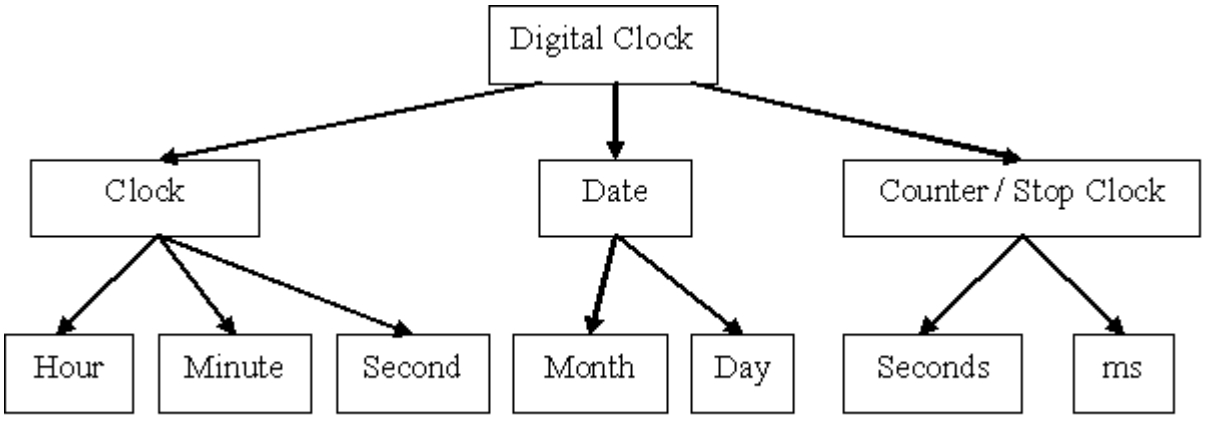
2.2.3 Time Module

The time module is a built-in Python module that provides functions for working with dates and times. We will use the time module to get the current time and update the clock in real-time.

The time module will provide the following functions:

- time.strftime: Returns the current time in a specified format.
- time.localtime: Returns the current local time.
- time.sleep: Pauses the program for a specified amount of time.

2.3 Architecture Diagram



CHAPTER 3

MODULES

3.1 Time Module

The time module in Python provides functions for working with time, primarily focusing on measuring time durations, controlling program execution flow, and converting between different time representations. It is essential for tasks like benchmarking code performance, scheduling events, and pausing scripts. The module operates mainly with two concepts of time: the epoch and struct_time. The epoch is the point where time begins for a given operating system (for Unix systems, it's January 1, 1970, 00:00:00 UTC), and time is measured in seconds since this epoch. The time.time() function is the most fundamental, returning the current time as a floating-point number representing seconds since the epoch. This is crucial for calculating the duration of an operation by taking a timestamp before and after the operation and finding the difference. Another key function is time.sleep(seconds), which suspends the execution of the current thread for the specified number of seconds, making it indispensable for controlling the rate of a program, such as in polling loops or setting up delays in a digital clock simulation.

The time module also handles the conversion of seconds into a more human-readable format known as struct_time. This is a named tuple that contains nine elements: year, month, day, hour, minute, second, weekday, day of the year, and Daylight Saving Time (DST) flag. The function time.localtime() converts a time in seconds since the epoch into a struct_time object representing the local time, taking into account the system's timezone. Conversely, time.gmtime() does the same but expresses the time in Coordinated Universal Time (UTC). For converting a struct_time object back into seconds since the epoch, the time.mktime() function is used. Finally, for formatting the struct_time object into

a custom string, the time.strftime() function (which shares a name with a function in the separate datetime module) is employed. This function uses format codes (e.g., %Y for year, %H for hour) to create a specific output string, which is often the final step in displaying time information to a user. While the datetime module often provides a more robust and object-oriented approach to date and time manipulation, the time module remains critical for low-level system timing and precise program flow control.

3.2 Tkinter Module

The tkinter module is Python's standard GUI (Graphical User Interface) toolkit. It provides a simple and fast way to create desktop applications with graphical elements like windows, buttons, text fields, and labels, making it highly relevant for creating the visual interface of a digital clock application. Tkinter is a wrapper around the Tcl/Tk toolkit, making it platform-independent and readily available with most standard Python installations. The process of building a GUI starts with the root window, which is instantiated using Tk(). This root window acts as the main container for all other elements, known as widgets.

Core to tkinter are its fundamental widgets, including Label (for displaying text or images, perfect for showing the time), Button (for interactive elements like setting an alarm), and Entry (for accepting single-line text input, such as entering a desired alarm time). All widgets are organized within a window using geometry managers. The three primary managers are pack(), grid(), and place(). The pack() manager organizes widgets in blocks, grid() places them in a table-like structure of rows and columns (excellent for precise clock layout), and place() allows positioning widgets using absolute coordinates.

3.3 Date Time Module

The datetime module provides classes for manipulating dates and times in both simple and complex ways. Unlike the time module, which focuses on timestamps and C-level structure, datetime is object-oriented, providing classes that make handling date and time arithmetic intuitive and readable. The module defines several key classes: date, time, datetime, and timedelta. The datetime class is the most powerful, combining both date (year, month, day) and time (hour, minute, second, microsecond) information into a single object. Creating a datetime object is simple: datetime.datetime(year, month, day, hour, minute, second). The function datetime.datetime.now() returns a datetime object representing the current local date and time, which is the primary source of time information for a digital clock application.

3.4 TTK Module

The ttk (Themed Tkinter) module is a critical enhancement to the basic tkinter library, designed to significantly improve the visual appearance and cross-platform consistency of Python GUIs. While the standard tkinter widgets often render with an archaic, platform-inconsistent look—appearing different on Windows, macOS, and Linux—ttk introduces a new set of themed widgets that automatically adopt the native look and feel of the user's operating system. This solves the problem of applications looking out of place or dated, providing a more professional and modern user experience for applications like a digital clock interface.

3.5 Strftime Function Module

It is important to clarify that strftime() is not a standalone module in Python; rather, it is a method (a function belonging to an object) found within both the

time module and the datetime module. The function's name stands for "string format time," and its sole purpose is to convert a time or date object into a custom-formatted string based on a set of standardized format codes.

3.6 OS Module

The os module provides a way of interacting with the operating system on which the Python code is running. This module is essential for performing platform-dependent tasks, such as file system operations, managing environment variables, and running system commands. The primary goal of the os module is to provide a consistent, high-level interface to the operating system's functions, ensuring that the same Python code runs correctly on Windows, macOS, and Linux without modification.

3.7 SYS Module (System-Specific Parameters and Functions)

The sys module provides access to system-specific parameters and functions and is tightly intertwined with the Python interpreter itself. It offers control over and insight into the environment in which the Python program is currently running. This module is crucial for tasks like handling command-line arguments, managing the interpreter's search path for modules, and gracefully exiting the program.

One of the most common uses of sys is accessing command-line arguments through the sys.argv list. This list contains all arguments passed to the script when it was executed, with sys.argv[0] being the name of the script itself. For a digital clock, this could be used to allow users to specify a startup option, like python clock.py -alarm-only. Another key feature is the management of the module search path via sys.path, which is a list of directory names where the interpreter

looks for modules when an import statement is executed. Developers can dynamically modify this list to include custom directories where their application's components are stored.

CHAPTER 4

CONCLUSION & FUTURE SCOPE

4.1CONCLUSION

Python as a Foundational Tool for Digital Clock Systems The exercise of modeling a digital clock's core problems using Python basics demonstrates that the language is an ideal foundational tool for both understanding and rapidly prototyping timekeeping and interface solutions. By employing simple data structures and control flow, we can effectively simulate complex real-world issues. The use of arithmetic operators and conditional loops in Python successfully models time drift (ϵ) and the logic required for external synchronization. The tkinter and ttk modules provide a direct and accessible pathway for building the GUI, addressing the low readability by enabling dynamic text formatting and modern theming. Furthermore, dictionaries and string manipulation are essential for simulating the remote command parsing necessary for connectivity (Problem 3), illustrating that the transition from a standalone device to a networked IoT component is fundamentally a software-logic problem. Finally, leveraging custom classes and lists of dictionaries in Python allows for the easy expansion of features (Problem 4), proving the scalability of a Python-based design from a simple clock to a sophisticated information hub. In essence, Python provides a powerful, readable, and highly reusable logic layer that can be easily transitioned to embedded systems or more complex frameworks.

4.2FUTURE SCOPE

The future scope for a digital clock project built on Python basics extends far beyond a simple display of time, focusing on deeper integration, intelligence, and system-level refinement.

4.2.1 Integration and Networking

Full IoT Implementation: Transition the simulated command parsing logic to use actual networking libraries like socket or asyncio to implement a functional Network Time Protocol (NTP) client and a simple web server. This allows the clock to communicate with other devices and truly solve the time drift problem through autonomous, high-accuracy synchronization.

4.2.2 Advanced User Experience and Hardware Control

Persistent Configuration: Employ the json or configparser modules to save user preferences, alarm settings, and display themes persistently across restarts, moving beyond simple in-memory variables.

4.2.3 Intelligence and Automation

Smart Alarms: Implement more complex logic involving the datetime module to create "smart alarms." These alarms could use conditional logic based on local traffic conditions (via an external API) or use the calendar module to avoid waking the user on public holidays, significantly enhancing the utility and feature set.

APPENDICES

APPENDIX A-SOURCE CODE

```
import time
import datetime

# ------------------ WEATHER (User Input) ------------------
city = input("Enter your city name: ")
temperature = input("Enter today's temperature (°C): ")
weather = input("Enter weather condition (Sunny/Cloudy/Rainy): ")

# ------------------ ALARM SETUP ------------------
alarm_time = input("Set Alarm (HH:MM:SS): ")

# ------------------ STOPWATCH ------------------
def stopwatch():
    print("\nStopwatch Started…")
    seconds = 0
    try:
        while True:
            mins = seconds // 60
            secs = seconds % 60
            print(f"Stopwatch: {mins:02d}:{secs:02d}", end="\r")
            time.sleep(1)
            seconds += 1
    except KeyboardInterrupt:
        print("\nStopwatch Stopped.\n")

# ------------------ MAIN CLOCK LOOP ------------------
print("\n--- DIGITAL CLOCK STARTED ---\n")
while True:
```

```
# Clock
current_time = time.strftime("%H:%M:%S")
current_date = time.strftime("%d-%m-%Y")

# Display Clock + Date + Weather
print("Time :", current_time)
print("Date :", current_date)
print("City :", city)
print("Temp :", temperature, "°C")
print("Weather :", weather)
print("----------------------")

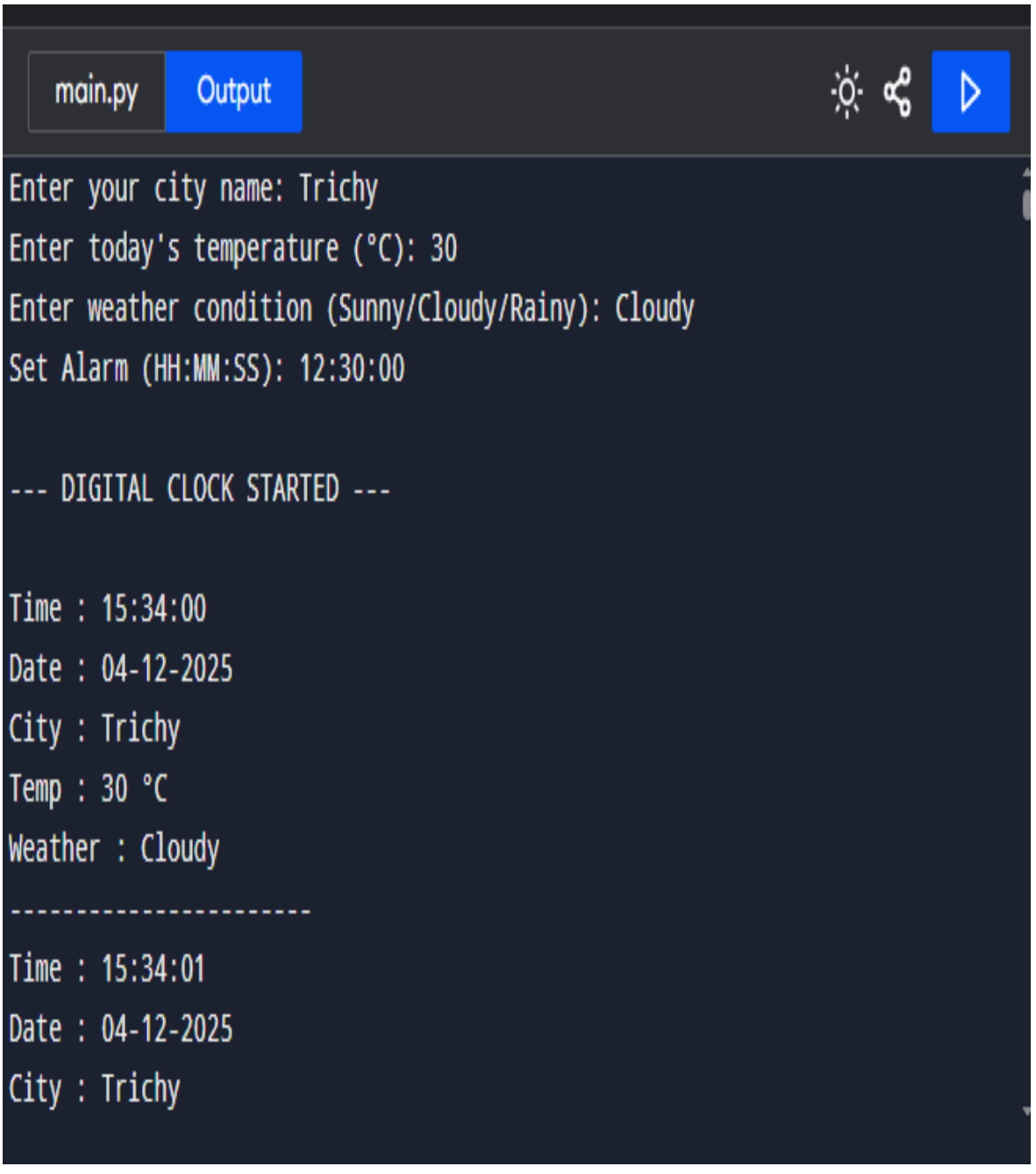
# Alarm check
if current_time == alarm_time:
    print("\n ALARM! Time reached! \n")

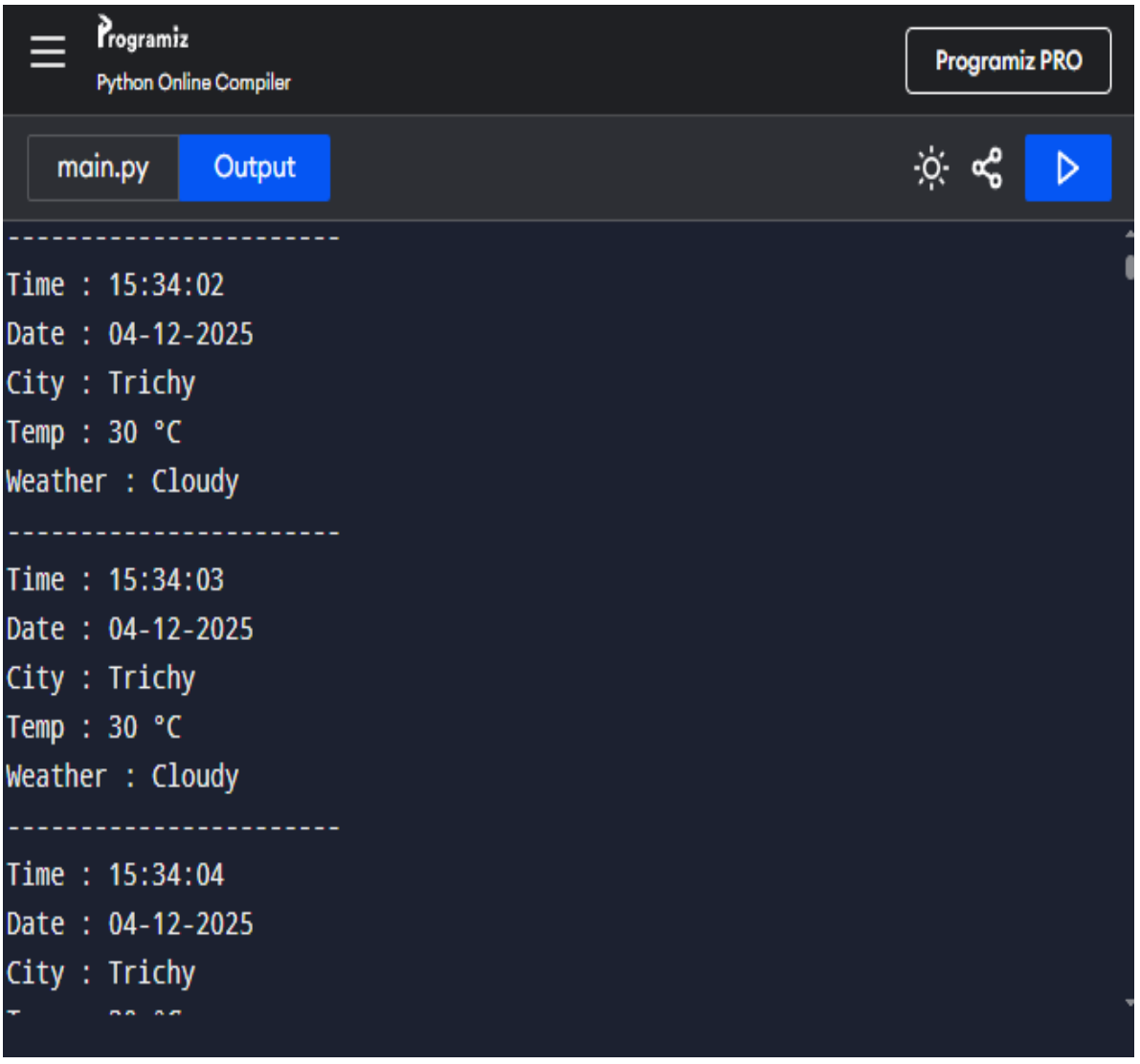
# Sleep for 1 second
time.sleep(1)

# Let user start stopwatch by pressing Ctrl+C
try:
    pass
except KeyboardInterrupt:
    stopwatch()
```

APPENDIX B SCREENSHOTS

RESULT AND DISCUSSION




Programiz
Python Online Compiler
Programiz PRO
main.py
Output

Time : 15:34:02
Date : 04-12-2025
City : Trichy
Temp : 30 °C
Weather : Cloudy

Time : 15:34:03
Date : 04-12-2025
City : Trichy
Temp : 30 °C
Weather : Cloudy

Time : 15:34:04
Date : 04-12-2025
City : Trichy

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This book provides comprehensive coverage of CRM concepts, strategies, and technological implementations.

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